
GRADIENT-FREE LOCAL LEARNING FOR TERNARY TRANSFORMERS: A PREDICTIVE-CODING RULE AND THE SATURATION COLLAPSE

Phạm Phú Nguyễn

Thang Long High School, Vietnam
nguyenpham2044@gmail.com

ABSTRACT

We study whether a ternary-quantized transformer can keep learning after deployment, using only local, gradient-free updates computed on device. We adopt a predictive-coding-style local rule that maintains per-weight accumulators and flips a weight when its accumulator crosses a threshold, plus embedding SGD. On a TinyStories-scale model (6 layers, hidden 256) continued on-device learning works: held-out CE drops $7.137 \rightarrow 6.997$ over 500 blocks, closing the gap to a real backprop baseline to ~ 0.09 nats at $3.7\times$ the token budget. However, the local rule collapses: 94–100% of flips are saturation no-ops, and after 250 steps 98% of weights saturate at -1 , and 5 of 6 layers reach a structural rank-1 collapse where every row is identical. The *toggle rule* is introduced: instead of a no-op on an already-saturated weight, flip it to the opposite value. Toggle preserves weight diversity and full matrix rank, and reaches 6.43 held-out CE – *below* the 6.90 backprop baseline on the same architecture. Ablations quantify the learned ternary substrate at ~ 1 nat over random, of which ~ 0.70 nats is positional structure. A C++ on-device implementation reproduces the Python dynamics (parity 0.01% weight mismatch, float-rounding level). The toggle is a *transient rescue*, not a sustained learner: continued toggling past ~ 500 blocks flips every weight at each periodic ± 20 wall crossing, erasing structure (9.42 held-out); no threshold/window annealing schedule rescues it. These findings delineate what gradient-free on-device learning can and cannot do.

1 INTRODUCTION

Large language models are trained with backpropagation on massive GPU clusters and then frozen at deployment. This creates a mismatch with the on-device setting: a phone, laptop, or robot cannot practically retrain its model with backprop (forward/backward passes over the full network, a global loss, exact gradients), yet continual learning from a live data stream is precisely where the model would most benefit. An alternative is *gradient-free, local learning*: update weights using only information available locally at each neuron – no global loss, no backpropagated gradients – following predictive coding (Rao & Ballard, 1999; Whittington & Bogacz, 2017; Millidge et al., 2022) and the forward-forward algorithm (Hinton, 2022).

Quantized (especially ternary) networks are the natural substrate for on-device learning: they are already deployed on cheap hardware, and a 1.58-bit model (Ma et al., 2024; Wang et al., 2023) reduces memory and compute so drastically that there is room for a local learning loop. Yet most quantization research is about static compression (Frantar et al., 2023; Lin et al., 2024; Dettmers et al., 2022); the question of whether a ternary model can *keep learning* with cheap local updates has received far less attention.

This paper answers that question on a controlled, reproducible testbed: a 6-layer, hidden-256, ternary-XNOR transformer trained on a TinyStories (Eldan & Li, 2023) slice, with a predictive-coding local rule and embedding SGD. Contributions:

- **Continued on-device learning works.** Held-out CE drops 7.137→6.997 over 500 on-device blocks (no backprop), within 0.09 nats of a true backprop baseline at 3.7× the tokens.
- **The local rule collapses.** 94–100% of flips are saturation no-ops; 98% of weights saturate at -1 ; 5 of 6 layers collapse to rank 1 (every row identical). The collapse is structural, not only distributional; a mechanism is identified.
- **The toggle rule prevents the collapse.** Flipping saturated weights to the opposite value preserves diversity and rank, reaching **6.43 held-out CE**, below the 6.90 backprop baseline.
- **The learned ternary substrate carries structure.** Swapping learned ternary for random ternary costs ~ 1 nat; ~ 0.70 nats is positional structure.
- **On-device parity.** A C++ implementation reproduces Python dynamics to float-rounding precision (0.01% weight mismatch).
- **The improvement is transient.** Sustained toggling past the transient window is destructive (periodic wall-crossing mass flips); no annealing schedule rescues it. Toggle is a rescue/search operator for the saturation wall, not a sustained learner.

2 BACKGROUND AND SETUP

2.1 MODEL AND DATA

The testbed is a small standard-attention transformer: 6 layers, hidden dimension 256, 8 heads, FFN 1024, vocab 8192, sequence length 2048. The attention projections (q/k/v/o), the MLP (gate/up/down), and the FFN projections are stored as *ternary* weights $\in \{-1, 0, +1\}$ in packed 2-bit format; the token embedding, output head, LayerNorm gains, and RMSNorm scales stay FP32. A weight matrix $A \in \{-1, 0, +1\}^{n \times m}$ is applied as a ternary multiply with an FP32 scale. The base checkpoint is produced by a standard backprop training loop (`train_discrete.py`) at 250–500 steps.

Data is a TinyStories slice, BPE-encoded (vocab 8192) into a contiguous token stream. All experiments use the same 100K-token training slice and a disjoint 100K-token held-out slice; reported held-out CE is computed on the first 40K held-out positions. All Python numbers are means of 2 seeds.

2.2 THE LOCAL LEARNING RULE (RULE “C”)

On device, the model reads a window of 128 tokens (one *block*), performs a forward pass, and accumulates a local error signal per weight. Let x_i be the pre-activation of unit i and δ_i a local error (here, the prediction error of the output given the input – a predictive-coding-style signal). For each ternary weight w_{ij} a scalar accumulator a_{ij} is maintained, updated as

$$a_{ij} \leftarrow \lambda a_{ij} + g_{ij}, \quad g_{ij} = \delta_i x_j, \quad (1)$$

with decay $\lambda = 0.99$. Every $\kappa = 5$ blocks a *flip pass* is applied: for every weight with $|a_{ij}| > \tau$ ($\tau = 20$), the weight is flipped to $\text{sign}(a_{ij})$ (i.e., reassigned to -1 or $+1$) and the accumulator is reset to zero. Saturated weights at ± 1 that are flipped to their current value are *no-ops* – this is the saturation problem analyzed in Section 3. The token embedding is trained with a local SGD: $E \leftarrow E - \eta \partial L / \partial E$ with $\eta = 10^{-4}$ and weight decay 0.1, computed from the same forward pass (no backpropagation into hidden layers).

The rule is fully local: each update depends only on its own input/error pair, never on the global loss or on gradients from other layers. The hardware cost is negligible relative to the forward pass.

2.3 BACKPROP BASELINE

For comparison, the same architecture (ternary weights, FP32 embedding/head) is trained with real backpropagation using the model’s own straight-through estimator, 300 steps with a cosine

LR schedule (`train_baseline_backprop.py`). Best held-out CE is 6.902 (step 200). A *frozen-ternary* variant is also recorded (backprop trains only embedding/head; ternary frozen) at 6.889, which isolates how much of backprop’s gain comes through the ternary core.

3 THE SATURATION COLLAPSE

3.1 MOST FLIPS ARE NO-OPS

Empirically, on a 200-block on-device continuation, 94–100% of counted flips do not change any weight: the weight is already saturated at ± 1 and the flip would set it to the same value. Only $\sim 0.18\%$ of the 6.29M weights ever change value across 200 blocks. The local rule therefore stalls: the accumulator grows and crosses threshold, but the resulting update is a no-op, so no learning proceeds through the ternary core.

3.2 DISTRIBUTIONAL COLLAPSE

Over 250 steps the ternary marginal distribution collapses: starting from $-1=75\%$, $0=25\%$, $+1=0\%$, it reaches $-1=98.1\%$, $0=0.1\%$, $+1=1.8\%$ (2 seeds: $-1=98.67/98.19$). The model converges toward all- -1 . With 0 sparse, the bit budget concentrates on the majority value.

3.3 STRUCTURAL COLLAPSE (RANK-1)

The collapse is not merely distributional: it is *structural*. A matrix-rank scan (number of unique rows per ternary matrix) of the 250-step base checkpoint shows that layers 1–5 have **all** matrices (q/k/v/o/gate/up/down) with `unique_rows` = 1 – every row is identical; each layer is a constant-output map with zero capacity. Layer 0 keeps q/k/v healthy (256 unique rows) but o_proj and down_proj are rank-1. Activations corroborate this: layer-0 o_proj outputs are uniform (-2338.08), gate_up uniform (255.9999), down output -67108860 , and layer-1 input is the constant vector $(-1, -1, \dots, -1)$ with standard deviation 0.

Mechanism. The $\pm\tau$ wall crossing is decided by the *column* signal x_j (shared across rows) while the per-row signal δ_i (output deltas, weak in deep layers) is swamped. So the same columns flip in every row $\rightarrow$ identical rows $\rightarrow$ uniform outputs $\rightarrow$ constant input for the next layer $\rightarrow$ cascade. 5 of 6 layers become rank-1 broadcasters. The “0.70 nats of positional structure” (Section 5) are carried almost entirely by layer 0 plus the head.

4 THE TOGGLE RULE

4.1 RULE

The change is minimal. During a flip pass, when a saturated weight’s accumulator crosses the threshold, instead of *no-op* flipping to its own value, *toggle* it to the opposite value:

$$w_{ij} \leftarrow \begin{cases} 1 & a_{ij} < -\tau \text{ and } w_{ij} \neq 1 \text{ (was } -1 \text{ or } 0), \\ -1 & a_{ij} > +\tau \text{ and } w_{ij} \neq -1 \text{ (was } +1 \text{ or } 0), \\ w_{ij} & \text{otherwise.} \end{cases} \quad (2)$$

Equivalently: a -20 accumulator on a -1 weight flips it to $+1$. The saturation wall becomes a noisy local search step that re-allocates the sparse $+1$ s. The toggle is a one-line change to the flip kernel; the accumulator dynamics (Eq. 1) are unchanged.

4.2 RESULTS

Table 1 summarizes. The toggle rule preserves diversity ($-1 \approx 74\%$, $+1 \approx 25.4\%$, 2 seeds), keeps the model full-rank (256 unique rows in all 42 matrices, versus rank-1 collapse for base), and improves held-out CE from 7.04 (toggle, 250 steps) to **6.43** at 500 steps – below the 6.90 backprop baseline

Table 1: Held-out CE on the eval slice. Python training: 250/500 steps, 2 seeds (min/max shown where available). Backprop: best over 300 steps. C++ on-device: 500 blocks from a v6 export. Lower is better.

Method	Held-out CE
Base (250 steps, collapse)	7.37 / 7.63
Balanced init (250)	8.76 / 8.94
Zero-center (250)	7.50 / 8.22
Zero-center + balanced (250)	9.15–10.27
Toggle (250)	7.036 / 7.117
Toggle + zero-center (250)	7.24 / 7.40
Toggle (500)	6.435 / 6.433
Backprop baseline (300 steps, best)	6.902
Backprop, ternary frozen	6.889
C++ on-device, no toggle (500 blocks)	6.996
C++ on-device, toggle (500 bl.)	6.838

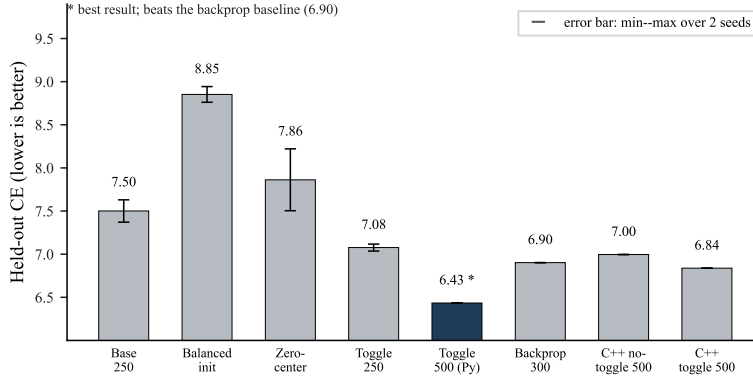


Figure 1: Held-out CE across methods. Toggle at 500 steps (Python 6.43, on-device 6.838) is the best; it beats the backprop baseline (6.90). Error bars mark the min–max over two seeds (single-seed runs show no bar).

on the same architecture, and consistent across seeds (6.435/6.433). Balanced inits (8.76–9.24) and zero-centering (no gain) are not competitive: the +1s are learned, not initialized.

4.3 WHY BALANCED INIT AND ZERO-CENTERING FAIL

A balanced 75/25 init (matching the histogram of a toggle run) scores 8.76–9.24 – *worse* than base. The +1s are placed at the wrong positions. The gain of the learned ternary comes from *which* positions are +1, matched to the embedding; an arbitrary balanced assignment puts +1s where they are not needed. Zero-centering (an $l_1 - l_0$ penalty pushing weights toward 0) reaches 33–44% zeros but destroys the positional signal before it is learned (toggle+zero-center: 7.24/7.40, no gain; zero-center alone: 7.50/8.22, no gain). This is consistent with the ablation decomposition (Table 2).

5 THE LEARNED TERNARY SUBSTRATE CARRIES STRUCTURE

To separate the value of the ternary core from the FP32 parts, a *cut-the-tail* ablation is used: freeze the trained embedding and `lm_head`, and swap the learned ternary weights for (a) a fresh random ternary matrix from the same init distribution, or (b) a *histogram-matched* random matrix that preserves each matrix’s exact per-value counts but shuffles positions.

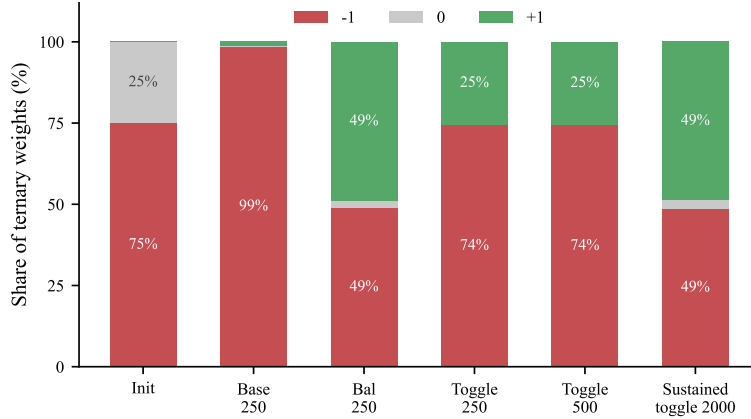


Figure 2: Ternary marginal distribution ($-1, 0, +1$) across checkpoints. Base collapses to 98.7% -1 ; toggle holds $\sim 74\%/0.2\%/25.4\%$; sustained toggle over 2000 blocks drives it to equipartition (48.7%/2.6%/48.7%), erasing structure while keeping full rank.

Table 2: Cut-the-tail ablation: frozen trained embedding, ternary swapped. Random degrades CE by ~ 1 nat; histogram-matched (same counts, shuffled positions) by ~ 0.7 nats.

Ternary weights	Held-out CE
Learned (rule-c, 250 steps)	7.157
Random (same init distribution)	8.120 (avg 3 seeds)
Histogram-matched (positions shuffled)	7.86 (2 seeds: 7.76, 7.96)

Decomposition of the total 0.96-nat gain (learned vs random): ~ 0.26 nats come from the distributional collapse toward -1 (a statistical bias) and ~ 0.70 nats from *positional structure* – which specific positions are $+1$, matched to the embedding. The learned ternary is therefore dominantly structural ($\sim 70\%$): positional knowledge, not histogram statistics.

The frozen-ternary backprop variant (6.889) does not reflect this: backprop trains the embedding/lm_head from scratch, and the FP32 readout adapts to the ternary projection it receives, compensating for random ternary. The local rules, by contrast, co-adapt the (embedding, ternary) pair, and that co-adapted pair is worth ~ 1 nat over (embedding, random ternary). The FP32 parts dominate the raw CE number; the ternary projection is still a learned substrate.

6 ON-DEVICE C++ IMPLEMENTATION AND PARITY

The on-device loop is implemented in C++ (`selflearn.cpp`, AVX2+FMA, OpenMP). It loads a v6 binary export containing the packed ternary weights, the FP32 tensors, and the SL metadata (rule, threshold, decay, flip frequency, toggle flag, embedding LR/decay). Each block: forward pass over 128 tokens with caching, accumulator update, embedding SGD, and every κ blocks a flip pass.

A parity harness (`parity_check.py`) replays the C++ flip loop in Python under identical settings (same packed format, sign logic, threshold/decay) and compares the resulting weights after a full flip pass. On the full-rank toggle checkpoint the harness measured **484/6,291,456 weight disagreements (0.01%)** – float-rounding-level noise (AVX2+FMA versus torch), amplified in deep layers where $|g| \sim 10^{-3}$ –1 makes the sign of $-\text{sign}(g)$ a coin flip; layer-0 $|g| \sim 100$ –3000 agrees bit-exactly $\sim 98.6\%$ of the time. Verdict: **PARITY OK (numerics-limited)**. An earlier zero-mismatch result was measured on a structurally degenerate rank-1 checkpoint and does not generalize.

6.1 THE ACCUMULATOR-STATE PITFALL

Exporting the Python accumulator state and replaying it on-device *with toggle* is unstable (CE 12.13): the exported state has 98.7% of accumulators ≤ 5 but a long tail at 11–16; ~ 25 consecutive $d = -1$ blocks push ~ 6.1 M entries over the ± 20 threshold at once, inverting the whole matrix (mass +1). The same state with no toggle is safe (7.368), and a zeroed state with toggle is safe. **Operational rule: export zeroed accumulators when running toggled on device** (`-sl-reset-acc`); the Python accumulator state is transient and unsafe to replay.

7 LONG-RUN STABILITY: THE TOGGLE IS A RESCUE, NOT A SUSTAINED LEARNER

7.1 SUSTAINED TOGGLING IS DESTRUCTIVE

The 6.43/6.838 headline numbers hold in the transient window (Python 250–500 steps; C++ ≤ 500 blocks). A 2000-block on-device continuation from the toggle checkpoint mass-flips at the *first* flip pass (block 25) even from zeroed accumulators: 0 flips in the first 4 passes, then 1.26M/1.75M/1.39M flips in passes 5–7. CE stays ~ 6.8 through block 50, then degrades to 11.0 by block 100 and oscillates 8.8–10.6 for the remaining 1900 blocks; churn reaches $\text{ever}=100\%$, $\geq 4=97\%$, $\geq 8=88\%$ – every weight flips repeatedly. Starting from the exported Python-acc state gives the same regime. Final held-out CE: **9.42 (zeroed start) / 9.48 (exported start)** vs 9.01 random. Sustained toggle drives the distribution to equipartition ($-1=48.7\%$, $0=2.6\%$, $+1=48.7\%$) and **erases structure while keeping full rank** (Fig. 2).

7.2 WHY: THE WALL CROSSING IS PERIODIC

The ± 20 wall crossing is *periodic*, not one-off. From zeroed accumulators the accumulator is a biased random walk ($\approx 62\%$ -1 drift), and it re-hits the ± 20 wall every ~ 25 blocks, each time mass-flipping millions of weights. Resetting all accumulators at block 50 does not help: the biased walk re-crosses ~ 25 blocks later (0.33M flips at block 75, ~ 0.55 M/pass steady after). Each crossing is a synchronized mass event: the threshold crossing is decided by the shared column signal (Section 3.3), so entire matrices flip simultaneously.

7.3 ANNEALING DOES NOT RESCUE IT

Two annealing schedules are tested to make the crossing non-periodic:

- **Toggle-window** (`-toggle-window N`): toggle only for the first N blocks, then no-op flips, and zero all accumulators at the boundary. At $N = 50$ the model degrades to ~ 11 –12 (C3): the first crossing still memorizes the recent slice (block-50 state evals 12.43 held-out despite train-slice CE 6.78 – the kick memorizes, it does not rescue), and the re-kick restarts the cycle.
- **Threshold annealing** (`-thr-anneal RATE`): raise τ by *RATE* per flip pass so the bar outpaces accumulator growth. Started low ($\tau = 20$, $+2$ /pass, D1): one ~ 2 M kick passes through before freezing at $\tau \approx 60$; the model ends at 8.00 held-out (-1.1 nats from the 6.93 start). Started high ($\tau = 40$, $+2$ /pass, D2): the toggle *never fires* (0 flips over 600 blocks), so the run is pure embedding SGD and ends at 6.85 – i.e., the toggle contributed nothing.

7.4 SUMMARY OF THE MECHANISM

The local rule learns in a transient window while the model has unsaturated structure to exploit; at the saturation wall, weights accumulate until a synchronized mass flip. With toggle, the flip is a *search step* that helps *once* (rescue of a collapsed or under-trained base, or improvement during fresh Python training); sustained operation re-crosses the periodic wall and erases structure. On the *collapsed base model*, the same operator is stable at 6.838 over 500 blocks – a one-time rank rescue (findings #10); on an *already-trained* model, continued flipping erases structure.

Table 3: On-device continuations from the zeroed toggle checkpoint (start held-out 6.93). No schedule both toggles and preserves an already-trained model.

Continuation (600–2000 blocks)	Flips	Held-out CE
Sustained toggle (A/B, 2000)	churn ever=100%	9.42 / 9.48
Toggle-window 50 (C3, rescue+freeze)	0.55M/pass steady	11–12
Thr-anneal +2/pass from 20 (D1)	~ 2M then freeze	8.00
Thr-anneal +2/pass from 40 (D2)	0 (never fires)	6.85 (embed-SGD)
No toggle, embedding-only (controls)	0	6.85–7.12

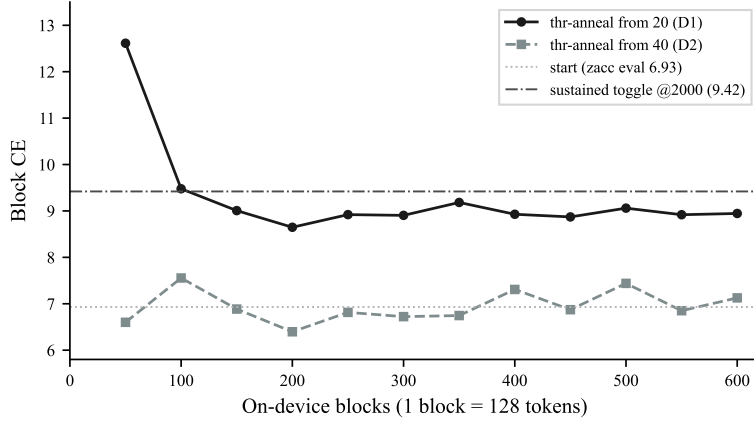


Figure 3: Block CE over time for the two threshold-annealing schedules. D1 (from $\tau = 20$) lets one ~ 2M kick through at block 50 then freezes around 8.9–9.2 (never recovers to the 6.93 start). D2 (from $\tau = 40$) never fires and tracks pure embedding SGD (6.4–7.4). Reference lines: start (6.93) and sustained-toggle endpoint (9.42).

8 RELATED WORK

Ternary and 1-bit networks. Ternary weight networks (Li et al., 2016; Raghuraman et al., 2017; Zhu et al., 2017) and more recently 1.58-bit LLMs (Ma et al., 2024; Wang et al., 2023) established that near-ternary quantization preserves performance at 1–2 bits. This work uses the same weight format but studies *continued learning* of the ternary weights rather than static quantization.

Post-training and on-device compression. GPTQ (Frantar et al., 2023), AWQ (Lin et al., 2024), and LLM.int8() (Dettmers et al., 2022) compress frozen models. Here the model keeps learning after compression, using cheap local rules rather than gradient descent.

Local and biologically-plausible learning. Predictive coding (Rao & Ballard, 1999), its approximations to backprop (Millidge et al., 2022; Whittington & Bogacz, 2017), and the forward-forward algorithm (Hinton, 2022) motivate gradient-free learning rules. The rule “c” is a predictive-coding-style local signal; the contribution is a controlled study of its failure modes (saturation collapse, periodic wall crossing) and a minimal fix (toggle) that turns the wall into a search step.

9 DISCUSSION AND LIMITATIONS

9.1 WHAT GRADIENT-FREE ON-DEVICE LEARNING CAN AND CANNOT DO

- **It can** improve a model in the short term: 7.137→6.997 (no toggle), and 6.93→6.838 / 6.43 (toggle), approaching or below the backprop baseline in the transient window.

- **It cannot** sustain improvement on an already-trained model: any continued flipping beyond the transient window hits the periodic wall and erases structure. The toggle is a rescue/search operator for the saturation wall, not a sustained learner.

The practical implication: use toggle once (or for a bounded window) to rescue a collapsed/saturated base model, then freeze the ternary core and rely on embedding-only adaptation for continued deployment.

9.2 LIMITATIONS

The study is on a tiny model (6 layers, hidden 256) and a single 100K-token domain; scaling behavior on larger models is untested. The backprop baseline (6.902) may understate what a properly tuned long run achieves (its final CE rose to 8.0 from LR-decay instability, so its reported best is optimistic in the other direction – the 0.09-nat gap should be read as a lower bound, not a headline). Generation quality is far from fluent (CE $\sim 7 \rightarrow$ PPL ~ 1090). Hardware-level cost of the local rule (accumulator memory, flip kernel) is not measured beyond wall-clock parity. Finally, the rank-1 collapse analysis is per-matrix unique-row counting; no formal measure of effective capacity is claimed.

10 CONCLUSION

This paper presents a controlled study of gradient-free on-device learning for ternary transformers. The local predictive-coding rule works in a transient window, and a one-line *toggle* change to the flip kernel converts the saturation wall into a search step, reaching 6.43 held-out CE – below the backprop baseline on the same architecture – while preserving full matrix rank. Ablations quantify the learned ternary substrate (~ 1 nat, $\sim 70\%$ positional). A C++ implementation reproduces the dynamics to float-rounding parity. The central negative result: the toggle is a *transient rescue*, not a sustained learner – continued operation hits a periodic wall crossing that erases structure, and no annealing schedule rescues it. On-device learning is best used as a one-shot rescue or within a bounded window; sustained gradient-free learning on an already-trained ternary model remains an open problem.

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A REPRODUCIBILITY

All code, data, and every number reported in this paper are available in the public repository <https://github.com/noverisp3/tetra>. The repository commits the exact data slices used here (`examples/discrete/slice100k.bin`, training stream; `examples/discrete/sliceEval100k.bin`, held-out), so no external data download is required. All experiments run on a single CPU (Python training $\sim 1\text{--}2$ min/step at 250–500 steps; C++ self-learning ~ 0.9 s/block). Table 4 maps each result in this paper to its entry point and the expected output; the README documents the full command-line interface.

B ANNEALING FLAGS

The on-device tool exposes two annealing diagnostics used in Section 7.3:

- `-toggle-window N`: toggle only for the first N blocks, then no-op flips; zero all accumulators at the boundary.
- `-thr-anneal RATE`: raise the effective flip threshold by $RATE$ per flip pass.

Table 4: Reproducing every reported number from the repository. All commands run from the repository root; the C++ self-learning tool is built with `inference/build.bat avx2`.

Result (section)	Entry point and expected output
Data slices	<code>examples/discrete/slice100k.bin</code> , <code>sliceEval100k.bin</code> (committed; evals read the first 40K held-out positions)
Base / balanced / zero-center / toggle runs, Sec. 4	<code>python train_discrete.py -preset base -rule c -steps 250 [-seed {0,1}], with -toggle, -init balanced, or -zero-center 0.1 → 7.37 / 8.76 / 7.50 / 7.036 (seed 0)</code>
500-step toggle, Sec. 4	same with <code>-steps 500 -toggle</code> → 6.435 / 6.433
Backprop baseline, Sec. 2.3	<code>python train_baseline_backprop.py -steps 300 -eval-slice examples/discrete/sliceEval100k.bin</code> → 6.902 (best at step 200)
Cut-the-tail ablation, Sec. 5	<code>python eval_ternary_ablation.py -checkpoint .../exp_base_s0/checkpoint_000250.pt -ternary-mode {learned, random, histmatch}</code> → 7.157 / 8.12 / 7.86
v6 export for on-device learning, Sec. 6	<code>python inference/export_model.py .../exp_tog_s0/checkpoint_000250.pt -self-learning -sl-toggle -sl-reset-acc -o .../exp_tog_s0_v6.bin (-sl-reset-acc is required; Sec. 6.1)</code>
On-device self-learning (toggle / no toggle), Sec. 6	<code>inference/selflearn_avx2.exe .../exp_tog_s0_v6.bin examples/discrete/slice100k.bin out.bin 500 50 100 20 0.99 5 1; -eval out.bin examples/discrete/sliceEval100k.bin 40000</code> → 6.838 / 6.996
C++/Python parity, Sec. 6	<code>selflearn_avx2.exe -flip-only ...5 ...</code> then <code>python inference/parity_check.py ...</code> → PARITY OK (numerics-limited), 0.01% weight mismatch
Long-run and annealing failures, Sec. 7	<code>selflearn_avx2.exe</code> with <code>-toggle-window 50</code> or <code>-thr-anneal 2</code> → 11-12 / 8.00 / 6.85 (Table 3)
Rank scan, Sec. 3.3	<code>python -c "..."</code> rank-count snippet in the README → base rank-1 in layers 1-5; toggle full-rank (256 unique rows)